

FS 572: Fermentation Science & Technology

Instructor: Jacob Graves

Lecture 4: Yeast Metabolism and Fermentation Kinetics

Learning Objectives

- Understand the role of *Saccharomyces cerevisiae* in industrial fermentation.
- Examine sugar utilization pathways and ethanol production.
- Evaluate the effects of temperature on fermentation performance.
- Identify common causes of stalled fermentations.

Key Concepts

Yeast converts fermentable sugars into ethanol and carbon dioxide through glycolysis and subsequent metabolic pathways. Fermentation rate is influenced by temperature, nutrient availability, oxygen exposure, and yeast health.

Discussion Questions

1. How does fermentation temperature affect flavor development?
2. What indicators suggest yeast stress?
3. Why is nutrient management important in high-gravity fermentations?

Lab Assignment

Record daily gravity readings for a controlled fermentation trial. Compare fermentation curves between two yeast strains and summarize results.